

DULUTH INTL, MN, USA

WMO: 727450

Lat: 46.837N

Lon: 92.183W

Elev: 437

StdP: 96.19

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14913

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.4	-24.5	-31.6	0.2	-26.8	-29.1	0.3	-23.7	11.6	-8.3	10.8	-9.8	4.6	310	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.1	21.0	27.5	19.7	25.9	18.7	22.5	27.3	21.2	25.8	20.0	24.2	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	16.3	25.4	19.6	15.1	24.2	18.4	14.0	22.7	68.4	27.1	63.5	25.7	59.3	24.2	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.6	8.6	-30.7	32.2	3.0	1.5	-32.9	33.3	-34.6	34.2	-36.3	35.0	-38.5	36.1
			-30.8	24.4	2.6	1.2	-32.7	25.2	-34.2	25.9	-35.7	26.6	-37.6	27.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-11.8	-9.5	-3.0	4.0	10.8	16.0	19.4	18.4	14.1	6.9	-1.1	-8.4
	DBStd	12.06	7.42	6.88	6.58	5.00	4.43	3.81	3.32	3.13	4.38	4.67	5.90	6.68
	HDD10.0	2896	674	547	403	188	43	3	0	0	12	121	333	571
	HDD18.3	5096	933	780	660	430	238	87	28	37	137	355	582	829
	CDD10.0	974	0	0	1	8	67	184	291	262	136	24	1	0
	CDD18.3	134	0	0	0	0	3	18	60	41	12	1	0	0
	CDH23.3	1069	0	0	0	2	56	175	482	276	74	4	0	0
	CDH26.7	226	0	0	0	0	11	34	115	56	9	0	0	0
(14) Wind	WSAvg	4.4	4.7	4.7	4.6	4.9	4.7	4.1	3.8	3.7	4.2	4.7	4.6	4.6
(15) Precipitation	PrecAvg	776	27	23	42	62	83	105	95	99	92	68	48	33
	PrecMax	1102	119	69	130	208	195	255	222	262	238	173	137	94
	PrecMin	504	3	3	6	6	4	14	21	18	21	12	5	3
	PrecStd	125	22	15	28	36	38	43	44	52	47	40	33	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	7.1	16.0	21.6	28.2	30.2	31.7	30.7	27.9	23.3	16.0	5.5
		MCWB	1.9	3.8	11.5	11.5	17.0	21.9	23.1	22.4	20.1	15.2	10.0	2.5
	2%	DB	1.8	4.0	11.3	18.0	24.7	27.3	29.5	28.1	25.5	19.4	11.9	3.2
		MCWB	0.5	1.9	6.8	9.8	15.0	19.1	21.4	20.7	19.0	13.5	8.7	1.6
	5%	DB	0.0	1.9	8.0	14.9	21.7	25.3	27.9	26.3	23.3	16.5	9.0	1.5
		MCWB	-0.9	0.2	4.5	8.2	13.6	17.8	20.3	19.4	17.7	11.7	5.8	0.4
	10%	DB	-1.7	0.0	5.2	12.1	18.9	23.4	26.2	24.6	21.2	14.1	6.8	0.3
		MCWB	-2.6	-1.3	2.6	6.4	12.2	16.7	19.3	18.5	16.6	10.3	4.5	-0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	4.3	12.2	13.8	19.1	23.3	24.3	24.0	21.8	17.0	11.8	3.7
		MCDB	4.3	6.4	14.5	18.3	25.0	28.1	29.9	27.8	26.2	21.5	14.1	4.4
	2%	WB	0.6	2.1	7.5	10.5	16.6	20.7	22.8	22.1	20.2	14.7	8.7	1.7
		MCDB	1.7	3.6	10.6	16.7	21.9	25.9	27.8	26.7	23.8	17.8	11.6	2.8
	5%	WB	-0.9	0.4	4.8	8.6	14.8	19.1	21.5	20.7	18.8	12.6	6.4	0.6
		MCDB	0.0	1.7	7.5	14.5	20.1	23.1	26.1	24.6	22.2	16.1	8.5	1.5
	10%	WB	-2.6	-1.3	2.8	6.9	13.0	17.6	20.2	19.5	17.3	10.7	4.6	-0.6
		MCDB	-1.7	0.1	5.1	11.6	18.2	22.0	24.7	23.3	20.3	13.5	6.7	0.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.2	8.9	9.5	10.8	10.7	10.5	10.0	9.6	8.1	7.1	7.2
		MCDBR	8.0	9.5	11.2	14.6	15.2	13.5	12.4	11.8	11.8	12.0	10.2	7.5
	5% WB	MCWBR	6.9	7.5	7.5	8.7	8.0	7.1	6.3	5.9	6.6	7.3	7.0	6.2
		MCDWR	7.8	8.0	10.5	13.6	13.3	12.0	11.4	10.5	10.2	10.7	9.7	6.8
(40) Clear-Sky Solar Irradiance	taub	taub	0.241	0.259	0.303	0.341	0.368	0.366	0.374	0.376	0.353	0.307	0.282	0.259
		taud	2.412	2.375	2.377	2.392	2.366	2.398	2.395	2.415	2.452	2.572	2.528	2.427
	Ebn at Noon	Ebn at Noon	878	926	925	915	897	898	885	869	860	863	820	808
		Edh at Noon	66	86	101	110	118	115	114	107	94	72	61	59
(44) All-Sky Solar Radiation	RadAvg	1.45	2.54	3.70	4.62	5.32	5.75	6.11	5.13	3.85	2.34	1.39	1.03	
	RadStd	0.16	0.23	0.35	0.46	0.39	0.40	0.25	0.37	0.27	0.30	0.16	0.10	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	+30
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page